

**MSTAR PRACTICE WORKSHEET****Lesson 6-3: Explore Numerical Expressions with Exponents**

Grade 6 Mathematics · Standard 6.AT.5

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

**Part A — Skills Check**

One point each.

**Question 1**    **SELECTED RESPONSE — CORRECT: D**What is the value of  $4^3$ ?

- ☐ A 7
- ☐ B 12
- ☐ C 43
- ☒ D 64

$$4^3 = 4 \times 4 \times 4 = 64.$$

**Question 2**    **SELECTED RESPONSE — CORRECT: C**What is the value of  $6^2$ ?

- ☐ A 8
- ☐ B 12
- ☒ C 36
- ☐ D 62

$$6^2 = 6 \times 6 = 36.$$

**Question 3**    SELECTED RESPONSE — CORRECT: B

Which expression shows  $5^3$  as repeated multiplication?

- ☐ A  $3 \times 3 \times 3 \times 3 \times 3$
- ☒ B  $5 \times 5 \times 5$
- ☐ C  $5 \times 3$
- ☐ D  $5 + 5 + 5$

$5^3$  means use 5 as a factor 3 times:  $5 \times 5 \times 5$ . The exponent tells how many times to multiply, not what to multiply by.

**Question 4**    SELECTED RESPONSE — CORRECT: C

Evaluate:  $3 \times (4 + 2)^2$

- ☐ A 36
- ☐ B 54
- ☒ C 108
- ☐ D 144

$4 + 2 = 6$ .  $6^2 = 36$ .  $3 \times 36 = 108$ .

**Question 5**    SELECTED RESPONSE — CORRECT: D

The drum loop plays  $3^4$  beats. What is the value of  $3^4$ ?

- ☐ A 7
- ☐ B 12
- ☐ C 27
- ☒ D 81

$3^4$  means use 3 as a factor 4 times:  $3 \times 3 \times 3 \times 3 = 81$ . It is NOT  $3 \times 4 = 12$ .

**Question 6**

SELECTED RESPONSE — CORRECT: B

A synth pattern repeats  $7 \times 7 \times 7$ . How do you write that as a power?

- ☐ A  $3^7$
- ☒ B  $7^3$
- ☐ C  $7 \times 3$
- ☐ D  $21^3$

The base 7 is used as a factor 3 times, so the power is  $7^3$ .  $7^3 = 7 \times 7 \times 7 = 343$ .

**Part B — Test-Format Questions**

Scoring notes and distractor rationales below each item.

**Question 7**

TWO-PART QUESTION

**Part A — correct answer: D**

Which value is equal to the power  $3^4$ ?

- ☐ A 7
- ☐ B 12
- ☐ C 64
- ☒ D 81

The base is 3 and the exponent is 4, so 3 is used as a factor four times:  $3 \times 3 \times 3 \times 3 = 81$ .

- **A:** That adds 3 and 4. The exponent tells you to multiply 3 by itself four times.
- **B:** That multiplies the base by the exponent. An exponent counts how many times 3 is used as a factor.
- **C:** That is four used three times, with base and exponent swapped. Here the base 3 is used four times.

**Part B — correct answer: C**

Which reasoning shows why your answer to Part A is correct?

- ☐ A Multiply the base by the exponent:  $3 \times 4 = 12$ .
- ☐ B Add the base and the exponent:  $3 + 4 = 7$ .
- ☒ C The exponent 4 tells how many times to use the base 3 as a factor:  $3 \times 3 \times 3 \times 3 = 81$ .
- ☐ D Use the exponent as the base and the base as the exponent:  $4 \times 4 \times 4 = 64$ .

An exponent counts repeated factors of the base. It does not tell you to multiply, swap, or add the two numbers.

**Question 8**

SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the power  $2^5$ .

- ☒ A The base is 2.
- ☒ B The exponent is 5.
- ☒ C It means  $2 \times 2 \times 2 \times 2 \times 2$ .
- ☐ D Its value is 10.
- ☒ E Its value is 32.
- ☐ F It means  $5 \times 5$ .

In  $2^5$  the base 2 is used as a factor 5 times:  $2 \times 2 \times 2 \times 2 \times 2 = 32$ . A value of 10 comes from multiplying  $2 \times 5$ , and  $5 \times 5$  swaps the base and the exponent.

**Part C — Show Your Thinking**

Model answer and scoring guidance.

**Question 9**

WRITTEN RESPONSE · 2 POINTS

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

**Model answer:** Doubling every hour means multiplying by 2 once per hour, so after 6 hours the count is  $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$  bacteria. Adding 2 each hour instead would give only  $1 + 12 = 13$ . Repeated multiplication compounds — each hour multiplies everything already there — while repeated addition only ever adds the same fixed amount.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.